

<https://doi.org/10.1038/s41524-024-01346-y>

Exploring superionic conduction in lithium oxyhalide solid electrolytes considering composition and structural factors

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Lithium (Li) oxyhalides have emerged as promising solid electrolyte candidates for all-solid-state batteries (ASSBs) due to their superior ionic conductivity and excellent cathode capability. However, the mechanism of Li transport in oxyhalides has remained unclear due to the complex nature of these compounds, which often comprise multiple phases such as crystalline and amorphous structures. Herein, a first-principles study using density functional theory and ab initio molecular dynamic simulation is performed to gain a theoretical insight into the structural behavior and conduction pathway in oxyhalide electrolytes. $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ electrolyte, as a case of study, is comprehensively investigated by considering various phases. It's revealed that the $\text{P}\bar{3}\text{m}1$ phase is energetically more stable and exhibits one order of magnitude higher Li conductivity compared to the C2/m phase among potential crystalline phases (1.63 mS cm^{-1} versus 0.24 mS cm^{-1} at room temperature). Interestingly, an amorphous phase with even higher ionic conductivity ($\sim 40 \text{ mS cm}^{-1}$) can be generated by creating LiCl-deficient Li-Zr-O-Cl compounds. The remarkably high ionic conductivity suggests that the amorphous structure is likely the dominant conducting pathway, facilitating fast Li transport due the weak bonding between Li and other atoms. Cl anion mobility is also observed in the amorphous phase through mean square displacement (MSD) calculation, although further analyses revealed this to be localized vibration. This study sheds light on the nature of oxychloride structures and promotes a deeper understanding of the superionic conduction mechanism in oxychlorides, which will contribute to the development of advanced solid electrolytes and ASSBs.

The growing demand of all-solid-state batteries (ASSBs) in electric vehicle and large-scale energy storage could be an important issue for the next generation economy^{1,2}. Solid electrolytes (SEs) are the key components to energy-density ASSBs by enabling Li metal anode and cathodes of high voltage^{3,4}. SEs feature stable performances over a wide temperature range and are safer than conventional liquid electrolytes^{5,6}. An ideal SE may have to meet many key characteristics, however ionic conductivity, electrochemical stability including Li-metal anode stability, and processibility are nowadays the most attractive properties studied for ASSBs, in which some electrolytes have shown excellent ionic conductivities and wide electrochemical stability and some are stable with Li metal anode^{3,7–9}.

A wide range of SEs including sulfides with the highest ionic conductivity but low electrochemical stability, oxides with high stability but low ionic conductivity, and halides with high ionic conductivity and good electrochemical stability are extensively studied^{10–12}. Halide SEs

have become a subject of great scientific and commercial interest due to well-balanced performances in ionic conductivity, processibility, cathode compatibility and cost^{8,13,14}. A variety of Li-M-Cl type halide SEs were synthesized, which involve Li_3YCl_6 ¹⁵, $\text{Li}_3\text{Y}_{1-x}\text{In}_x\text{Cl}_6$ ¹⁶, $\text{Li}_{3-x}(\text{Zr}/\text{Hf})_x(\text{Y}/\text{Er}/\text{Yb}/\text{In})\text{Cl}_6$ ^{17,18}, Li_3InCl_6 ¹⁹, $\text{Li}_x\text{ScCl}_{3+x}$ ²⁰, $\text{Li}_2(\text{Sc}/\text{In})_{2/3}\text{Cl}_4$ ^{21,22}, $\text{Li}_{3-3x}\text{M}_{1+x}\text{Cl}_6$ ($\text{M} = \text{Tb}, \text{Dy}, \text{Ho}, \text{Y}, \text{Er}$ and Tm)²³, Li_2ZrCl_6 ^{24–26}, $\text{Li}_{0.5}\text{SmCl}_{3.5}$ ²⁷, etc. The measured ionic conductivities are very attractive, with most of them are in between the range of 0.5 and 2 mS cm^{-1} and are even competitive with that of many sulfides SEs. Developing halide SEs with higher ionic conductivities has been a major research direction as there is plenty of room for improvement. There have been enormous efforts focusing on doping or substitution of M in Li-M-Cl in order to optimize the concentration of Li^+ and vacancy and/or to modify the Li migration paths and the energy landscape, however the improvement in ionic conductivity is limited.

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Very recently, a mixed-anion strategy has been proposed and introduced into the development of new halide SEs, which triggers significant progress in enhancing ion conduction in halides. A representative work was performed by adding some portion of oxygen into the chloride anion sublattice of Li-M-Cl (M = Nb, Ta), producing new lithium-metal-oxihalide (Li-M-O-Cl) materials with a record-breaking ionic conductivity (up to 12.4 mS cm^{-1}) for halide SEs^{28,29}. Similarly, Li-Zr-O-Cl series were developed following the mixed-anion strategy. The ionic conductivity of Li-Zr-O-Cl ($1\text{--}3 \text{ mS cm}^{-1}$) is nearly an order of magnitude higher than that of the single-anion counterpart of Li-Zr-Cl ($\sim 0.3 \text{ mS cm}^{-1}$). Although the ionic conductivity is not as high as that of Li-Nb(Ta)-O-Cl, Li-Zr-O-Cl has a remarkable advantage in cost, which will contribute to practical large-scale application^{30,31}. Therefore, Li-Zr-O-Cl series have been recognized as a promising SE candidate for ASSBs. Further optimization of the structure of Li-Zr-O-Cl SEs is likely to introduce faster ion conduction and is highly anticipated, which, however, is extremely challenging for the time being as the dominating conducting structure of Li-Zr-O-Cl SEs hasn't been confirmed yet.

The phase or space group of a structure is generally interpreted from the characteristic peaks of X-ray or neutron diffraction. The reported Li-M-Cl structures have been identified with different space groups such as C2/m, P $\bar{3}$ m1, Fd-3m, and Pnma³². The substitution of O²⁻ at Cl⁻ in Li-Zr-Cl systems were reported with different phases, in which diffraction peaks of $x\text{Li}_2\text{O-MCl}_x$ and $\text{Li}_{2.22}\text{Zr}_{1.11}\text{Cl}_{5.33}\text{O}_{0.67}$ belong to LiCl phase^{33,34}. However, a series of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ (like $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$) structures were reported with P $\bar{3}$ m1 (phase I) or C2/m (phase II) phase, and coexistence of phase I and phase II^{30,31}. The reported XRD curves for $x\text{Li}_2\text{O-MCl}_x$, $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$, $\text{Li}_3\text{Zr}_{0.75}\text{OCl}_4$, and $\text{Li}_{2.22}\text{Zr}_{1.11}\text{Cl}_{5.33}\text{O}_{0.67}$ have almost similar peaks and pattern that could rise an ambiguity between these reports. $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ (possessed P $\bar{3}$ m1, and C2/m and coexistence of both phases) exhibited room temperature ionic conductivity of greater than 1 mS cm^{-1} , however, further Li and Cl were depleted in $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ to enhance ionic conductivity. Such type of structures with multiple crystalline phases with intrinsic vacancies are interesting, however creating 0.75LiCl more vacancies to enhance ionic conductivity make it even more attractive for further insights, whether it can maintain its crystallinity for ASSBs?

This work presents a comprehensive first-principles study using density functional theory (DFT) and ab initio molecular dynamic (AIMD) simulations to study crystal structural, electronic, phase/electrochemical stability and conduction mechanism of oxyhalide $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ with different phases. Analyses are performed on AIMD simulations to compute mean square displacement (MSD), ionic conductivity, pair correlation function (PCF), vibrational density of states (VDOS), and probability density function. The results indicate that the structures are thermodynamically unstable with the hull energies of 33 and 41 meV/atom, Li-ionic conductivities of 1.63 and 0.24 mS cm^{-1} with an activation energy of 0.29 and 0.36 eV exhibiting 3-dimensional conduction path for P $\bar{3}$ m1 and C2/m phases, respectively. Li and Cl deficient composition ($\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$) are also studied and the results revealed that it is highly unstable and melted during AIMD simulations. Surprisingly, the obtained amorphous phase exhibits much higher ionic conductivity which indicates potential existence of amorphous phase in the deficient sample and explains the enhanced ionic conductivity as observed experimentally.

Results

Crystal structure

The reported series of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ (LZCO) materials were mechan-chemically synthesized with multiple phases depending on oxygen content. The crystal structures of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ are composed of different sites, intrinsic vacancies, and partial occupancies, which makes these structures complex for theoretical study. LZCO with both P $\bar{3}$ m1 and C2/m phases were considered for the simulations. The P $\bar{3}$ m1 phase contains two different Li atoms, Li1 and Li2 occupying 6g and 6h sites, four different Zr atoms, Zr1, Zr2, Zr3 and Zr4 at 1a, 2d, 1b and 2d sites, three Cl and O atoms at three different 6i sites with partial occupancies, respectively. C2/m phase consists

of three different Li atoms Li1, Li2 and Li3 occupying 4h, 2d and 4g sites, one Zr atom at 2a sites while two Cl and O sharing sites at 8j and 4i sites respectively. The occupancy of each atom at each site is different and directly related with O content in every composition.

The phases P $\bar{3}$ m1 and C2/m of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ ($x = 0, 0.25, 0.5, 0.75, 1, 1.25, 1.50$ and 2) were simulated to extract their reflection of powder diffraction plotted in Fig. 1. With $x = 0$, Li_2ZrCl_6 for the phase P $\bar{3}$ m1 has the highest peak near 30° of 2θ and the second highest peak near 15° of 2θ , however the first highest peak starts to increase near 15° of 2θ while second highest peak is near 30° of 2θ for $x = 0.25$. The simulated pattern for $0.5 \leq x \leq 2$ similar pattern was observed with the first highest peak near 15° of 2θ and the second highest peak near 30° of 2θ , whereas the 3rd and 4th peaks observed at 42° and 50° of 2θ are same for all LZCO compositions (Fig. 1a). The pattern exhibits pure phase and isostructural with Li_2ZrCl_6 about the composition of O with $x \leq 0.25$ and is in agreement with the experimental observed XRD pattern (Supplementary Fig. 1)³¹. The simulated pattern for the C2/m phase for $0 \leq x \leq 0.75$ indicates no change in the peaks (Fig. 1b). The crystallographic data for $x = 0.50$ and $x = 0.75$ ($\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{2.75}\text{ZrCl}_{5.25}\text{O}_{0.75}$) for C2/m phase provided in the experiment is inconsistent with the chemical formula, which may be the reason for no change in peaks. At $x = 1$, the first highest peak for C2/m phase is observed near 15° of 2θ and the second highest peak near 30° of 2θ , suggesting a phase transition. At $x = 1.25$ there is change and increase in the first highest peak and decrease in the second highest peak, which may be attributed to the coexistence of P $\bar{3}$ m1 and C2/m phases. There is again change in the first and the second peaks for $x = 1.5$, and much decrease in the peaks between 10° and 20° of 2θ and again increase in peak near 20° of 2θ in C2/m phase (Fig. 1b). These simulated diffraction patterns indicate the discrepancy between the crystal structures in C2/m phase with the increasing oxygen doping content. The reflection of crystallinity and amorphous behavior for simulated XRD pattern of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ ($x = 0, 1$ and 2) are further analyzed. The selected structures indicated maximum crystallinity with negligible amorphous phase (Supplementary Fig. 2).

The structures of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ are complex with intrinsic vacancies or partial occupation at all sites. Energetically most stable structures are needed to compute the key properties of electrolyte like phase stability, thermodynamic stability, and ionic transport. To have detailed structural analyses, $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ with P $\bar{3}$ m1 and C2/m phases were considered for comprehensive study using experimental crystallographic information³¹. Supercell program³⁵ is useful tool which helps to generate all possible structures based on their site occupancies using permutations and combinations. To maintain the crystal charge neutrality and minimum number of atoms meeting the reported chemical formula, supercell program was used to compute electrostatic Coulomb energy. Electrostatic Coulomb energy of each structure is computed using valency and site occupation of each atom (Li^{1+} , Zr^{4+} , Cl^{-} , and O^{2-}). The experimentally reported $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ P $\bar{3}$ m1 phase have Li1, Li2, Cl1, Cl2, Cl3, O1, O2, O3, Zr1, Zr2, Zr3 and Zr4 with occupancy of 0.53, 0.62, 0.95, 0.923, 0.977, 0.05, 0.077, 0.023, 0.649, 0.565, 0.709 and 0.256, respectively, while C2/m phase having Li1, Li2, Li3, Cl1, Cl2, O1, O2, and Zr1 with occupancy of 0.58, 0.2, 0.77, 0.825, 0.90, 0.175, 0.10 and 1.00, respectively (Fig. 1c, g). A series of crystal structures were generated, and the electrostatically most stable (lowest Coulomb energy) structure were extracted (Fig. 1d, h) for both phases. To make crystal structure more reliable, a little deviation in sites for both phases was made (P $\bar{3}$ m1 phase having Li1, Li2, Cl1, Cl2, Cl3, O1, O2, O3, Zr1, Zr2, Zr3 and Zr4 with occupancy of 0.50, 0.75, 0.9167, 0.8334, 1.00, 0.0834, 0.1667, 0.00, 0.50, 0.50, 0.50 and 0.50, respectively, while C2/m phase having Li1, Li2, Li3, Cl1, Cl2, O1, O2, and Zr1 with occupancy of 0.25, 0.50, 0.75, 0.825, 0.90, 0.125, 0.0 and 1.00, respectively) and electrostatically most stable structure were obtained (Fig. 1e, i). Ten electrostatically lower energy structures from each phase were considered to get energetically most stable for which spin polarization calculations were carried out to get atomically and volumetrically relaxed structures (Fig. 1f, j). Coulomb energy for each series and phase is plotted in Supplementary Fig. 3. The ground state energy structure is used to compute optimized lattice parameters for each structure.

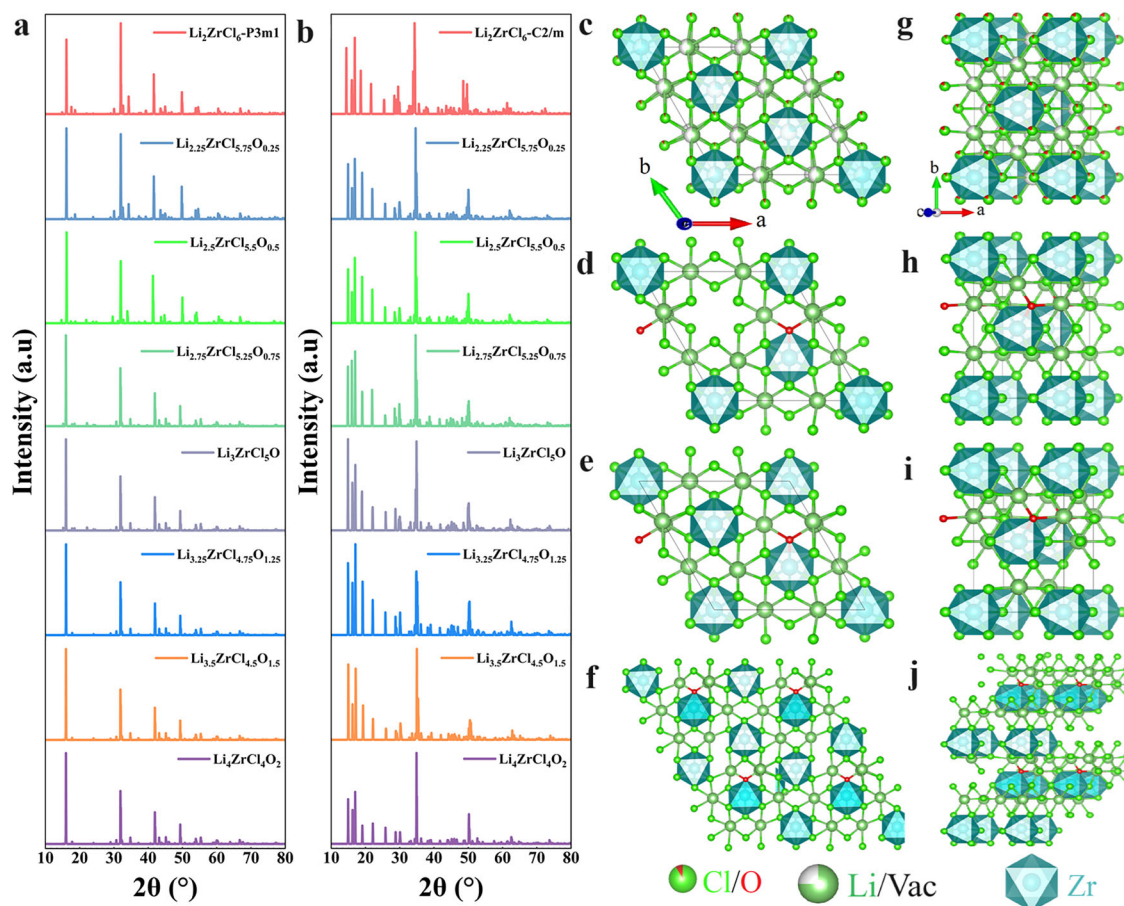


Fig. 1 | Simulated XRD patterns and electrostatically most stable structures. Simulated XRD patterns of $\text{Li}_{2+x}\text{ZrCl}_{6-x}\text{O}_x$ for phases with **a** space group $\text{P}\bar{3}\text{m}1$ and **b** space group $\text{C}2/\text{m}$. Crystal structure of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ with $\text{P}\bar{3}\text{m}1$ phase **c** with actual occupation of all atoms at each site, **d** electrostatically most stable structure with experimentally reported occupation sites, **e** electrostatically most stable structure with deviation in occupation sites, and **f** atomic and volumetric relaxed

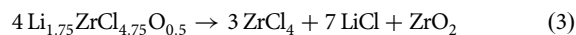
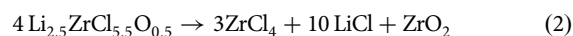
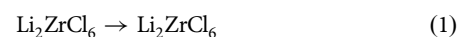
structure after spin polarization calculation. $\text{C}2/\text{m}$ phase **g** with actual occupation of all atoms at each site, **h** electrostatically most stable structure with experimentally reported occupation sites, **i** electrostatically most stable structure with deviation in occupation sites, and **j** atomic and volumetric relaxed structure after spin polarization calculation. Cl/O indicates O doping at Cl sites, Li/Vac indicates partial occupancies of Li ion at different sites with Li intrinsic vacancies.

Supplementary Fig. 4a indicates that the structure number 53 generated by supercell program has lowest electrostatic Coulomb energy (-311.564 eV), lowest total energy (-127.347 eV), and less hull energy (33 meV) computed by DFT calculation (Supplementary Table 3). These initial calculations are performed on the unit cell structure (28 atoms) with lattice parameter $a = b = 10.93700$ Å, $c = 6.02200$ Å for $\text{P}\bar{3}\text{m}1$ phase. The structure number 53, which has minimum ground state energy is used to compute optimal lattice parameters, Supplementary Fig. 4b indicates that the optimal volume is little lower than as compared to experimental reported lattice parameters. A similar process is used for $\text{C}2/\text{m}$ phase for all model structures (including LiCl deficient structure) used in this study. A supercell of $1 \times 1 \times 2$ with 56 atoms and $2 \times 1 \times 2$ supercell containing 76 atoms of $\text{P}\bar{3}\text{m}1$ and $\text{C}2/\text{m}$ phases respectively, whereas forty-eight (48 atoms) atomic model of $1 \times 1 \times 2$ of $\text{P}\bar{3}\text{m}1$ for amorphous structure was used for computing diffusion coefficients and ionic conductivity.

Phase stability and electrochemical window

The spin polarized calculations were performed to compute the stability, hull energy and electrochemical stability of the pure Li_2ZrCl_6 , $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$. It was found that both phases ($\text{P}\bar{3}\text{m}1$ and $\text{C}2/\text{m}$) of Li_2ZrCl_6 are thermodynamically stable with zero hull energy (equation 1), which is reflected by the ternary phase diagram at 0 K (Fig. 2a). However, oxygen containing structures ($\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$) are thermodynamically unstable with positive hull energy at 0 K and can be decomposed into other equilibrium phases according to

following reaction:



The structure of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ with $\text{P}\bar{3}\text{m}1$ is energetically more stable with hull energy of 33 meV/atom and 41 meV/atom for $\text{C}2/\text{m}$ phase, however both phases with O are thermodynamically unstable. Figure 2b shows the quarterly phase diagram of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ with green square in the inset green triangle indicating that it can be decomposed into the stable phases of ZrCl_4 , LiCl and ZrO_2 . Moreover, Li-depletion is an effective strategy to introduce structural variation and produce lattice defects for enhanced ionic transport in inorganic solid electrolytes. $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is a defect structure with the intrinsic depletion of $\text{Li}_{0.75}\text{Cl}_{0.75}$ in the $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$. However, the intrinsic deficiency of $\text{Li}_{0.75}\text{Cl}_{0.75}$ makes the structure even more unstable with the hull energy of 74 and 90 meV/atom in the $\text{P}\bar{3}\text{m}1$ and $\text{C}2/\text{m}$ phase, respectively. In general, such type of structures with hull energy more than 50 meV/atom are not easy to be synthesized but are accessible under certain circumstances^{36–38}. It is mentioned above that 300 crystal structures were generated for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ in this study by using the experimentally reported occupation of each atom, while 600

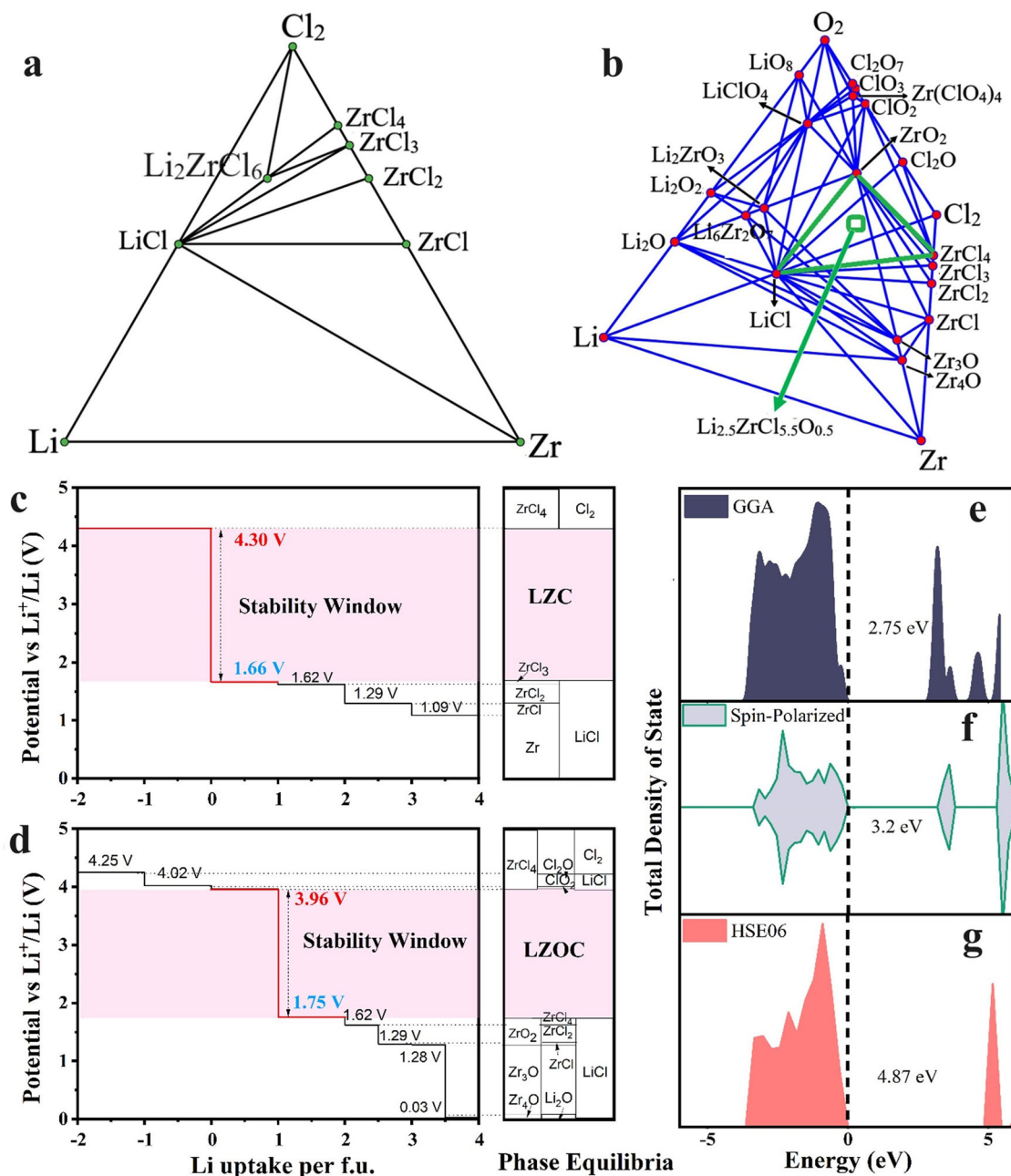


Fig. 2 | Phase diagrams, electrochemical stability window voltages and total density of states. Ternary and quaternary phase diagram constructed at 0 K for **a** Li_2ZrCl_6 and **b** $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$. Electrochemical stability window voltage for **c** Li_2ZrCl_6 , and **d** $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$. Total Density of states computed for

$\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ using **e** non-spin-polarized calculation with GGA functional, **f** spin-polarized calculation with GGA functional, and **g** non-spin-polarized calculation with HSE06 functional.

structures were extracted by deviation in occupation which exhibit low hull energy. However, no crystallographic information has been provided in the experimental study for deficient $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$. The modified structure information for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is provided in Supplementary Tables 1 and 2 of Supplementary note 1.

The electrochemical stability window against Li electrode for Li_2ZrCl_6 , $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ was investigated using Li grand potential phase diagram. Li_2ZrCl_6 exhibits a wide electrochemical stability window voltage (1.66–4.29 V versus Li/Li⁺) as compared to $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ (1.75–3.96 V) indicated by the green line in Fig. 2c, d, respectively. However, the results suggest that both Li_2ZrCl_6 and $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ are not stable with Li metal anode. The reduction and oxidation potentials and decomposition reactions into stable products are given in the Supplementary note 2. The

electrochemical stability window voltage of $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is same with that of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ (Supplementary Fig. 5) for P3m1 and C2/m phases.

The electronic property analysis was further carried out to compute bandgap for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ using spin, non-spin polarized calculations with GGA functional and HSE06 functional. Supplementary Fig. 6b shows that $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ is a large bandgap (≈ 5 eV) insulator that could be used for practical and dendrite free electrolytes for batteries^{39,40}, however the peaks of EDOS are not smooth. To make the EDOS smooth, Gaussian smearing method is used to extract the EDOS for GGA and HSE06 method with average number of configuration of 1000 and smearing value of 0.1 eV. The bandgap values computed by GGA functional (Fig. 2e, f) are much lower than bandgap of HSE06 functional (Fig. 2g), which results in the underestimation of GGA functional. Total and partial

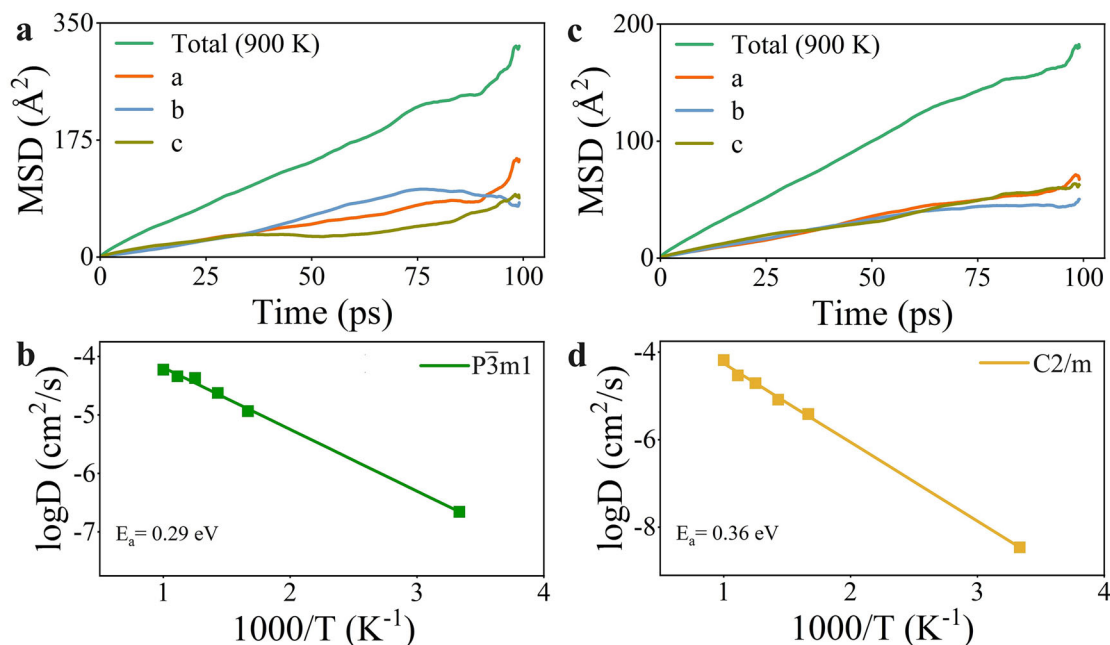


Fig. 3 | MSD and Arrhenius plots. Total and directional Li MSD extracted for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ at 900 K for **a** P3m1 phase and **b** C2/m phase. Arrhenius plots of diffusivity at different temperatures (600–1000 K) to compute extrapolated Li ionic conductivity at room temperature for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ **c** P3m1 phase, and **d** C2/m phase.

density of states (Supplementary Figs. 6 and 7) were also examined for both phases, with all the compositions exhibiting wide bandgaps (~ 5 eV).

Li-ion transport properties

To examine ionic transport, AIMD simulations were performed using NVT ensemble. Mean square displacement (MSD) and diffusion coefficients were obtained by the linear fitting of MSD from Vaspkit⁴¹ for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$. The $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ structure exhibits high Li MSD along three directions for both phases (Fig. 3a, c), which indicates fast three-dimensional diffusion of Li. The activation energy and Li ionic conductivity were examined by the Arrhenius plot of diffusivity (Fig. 3b, d). P3m1 phase of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ exhibits a high Li-ionic conductivity of 1.63 mS cm^{-1} with an activation energy of 0.29 eV, while C2/m phase exhibits a lower conductivity of 0.24 mS cm^{-1} with an higher activation energy of 0.36 eV, which is in agreement with experimentally reported ionic conductivity³¹. No diffusion of Zr, Cl and O was observed during AIMD simulations (Supplementary Fig. 8), suggesting that the $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ structure maintains its crystallinity at higher temperatures as well, whereas deficient $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ could not maintain its crystallinity at temperature as low as 600 K, and melted (Supplementary Fig. 9). These results show that $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is disordered structure with excess depletion of 0.75LiCl and makes it energetically unfavorable, which is investigated below in the amorphous part.

Pair correlation function (PCF) $g(r)$ of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ was extracted for more comprehensive and structural analysis from AIMD simulations and Li ion transport mechanism. The PCF between Li-Cl (Fig. 4a) and Li-O (Fig. 4b) is presented, which indicates that Li-O pair is the main contributor in highest peaks of PCF, suggesting, O is strongly arranged at their sites during Li diffusion. The first highest peak of Li-Cl is at the radius of 2.4 $\text{\AA}$, while Li-O first peak is at 1.9 $\text{\AA}$ at each simulation temperature for 100 ps. The second highest peak for Li-Cl and Li-O is at the radius of 4.6 $\text{\AA}$ and 4.4 $\text{\AA}$, respectively. The inset Fig. 4a, b correspond to the PCF results, where solid lines represent bond lengths associated with the first peaks, while dotted lines denote the second highest peaks and beyond. The height of the peaks for both pairs decrease with the increase in temperature from 600 K to 1000 K, which suggests enhanced Li diffusion with rise in simulation temperature. The PCF files between Li-Li, Li-Zr, Li-Cl, and Li-O for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ -C2/m phase was also calculated and presented in

Supplementary Fig. 10 at each temperature, which indicates similar behavior as observed in P3m1 phase. The Li-O pair has the maximum peak and decreased with the increasing temperature (Supplementary Fig. 11), suggesting high Li diffusion with increasing temperature.

VDOS was obtained from AIMD simulations versus frequency (eV) at temperatures of 600–1000 K for Li (Fig. 4c) and O (Fig. 4d) in $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ ^{42,43}. The VDOS of Li at each temperature shows the highest peak near frequency of 0.03 eV and becomes straight after 0.1 eV. There is a decrease in peak with increase in simulation temperature (Supplementary Fig. 12), which is again an indication of fast Li diffusion with rise in temperature. As a case study for immobile species, VDOS is computed for oxygen. It can be seen in Fig. 4d that the peak values are not normalized at a single frequency, indicating O is strongly bonded and resist the movement during AIMD simulations at each temperature. The result suggests that O remained localized at their mean position and no diffusion is observed for the immobile Zr, Cl and O, which confirms the MSD result during AIMD simulations.

Amorphous structure

In order to determine the formation of amorphous phase and ionic conduction mechanism, the $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ structure is simulated using P3m1 phase as an example. The structure was initially heated at 100 K for short simulation time, and heated at 300 K for 50 ps, at 600 K for 100 ps and finally heated at 1000 K for 100 ps to completely melt the structure. The structure was optimized after 1000 K with spin polarized DFT calculation. The optimized structure was finally quenched to 100 K with a constant rate of temperature gap of 200 K (800, 600, 400, 200 K) for 10 ps at each temperature using a time step of 2 fs with NVT ensemble using Nose'-Hoover thermostat⁴⁴. We have adopted the same method for the formation of amorphous just like other halide amorphous structures reported, including $\text{MAICl}_{2.5}\text{O}_{0.75}$ (M = Li, Na)⁴⁵, NaTaCl_6 ²⁷, LiTaCl_6 ⁴⁶, antiperovskite $\text{Na}_{3-x}\text{OH}_x\text{Cl}^{17}$. The complete melting, cooling and fully equilibrated process is plotted in Fig. 5a. AIMD simulations were carried out to heat the structure at 1000 K for 80 ps to completely melt the structure. It is a fact the structure shows diffusion for Cl, Zr and O at 600 K as well, however 1000 K is chosen so that the bonding of atoms, position of all atoms completely changes and melts. Finally, the structure was heated at different temperatures like 340, 360 and 380 K to get extrapolated ionic conductivity at 300 K. The structure

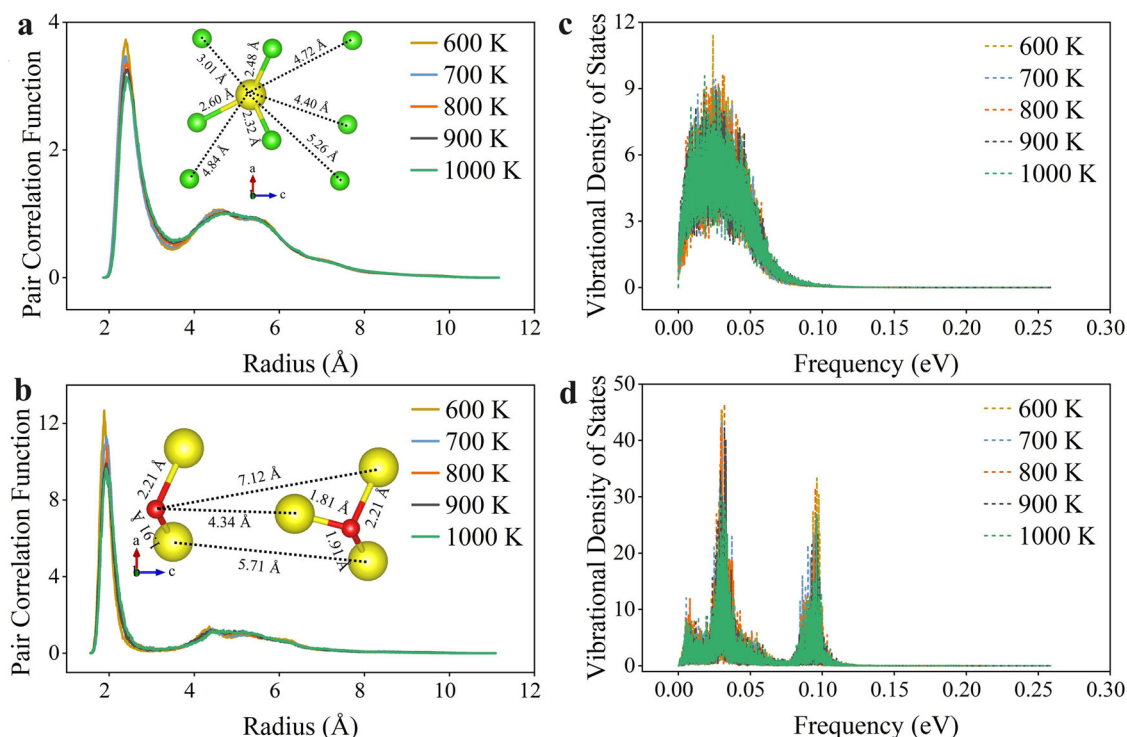


Fig. 4 | Pair correlation function and vibrational density of states. Pair correlation function extracted from AIMD simulations at each temperature (600–1000 K) for 100 ps of $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ for **a** Li–Cl pair and **b** Li–O pair. Vibrational density of

states (VDOS) obtained from AIMD simulations at each temperature (600–1000 K) for 100 ps of $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ of **c** Li and **d** O atom.

was also directly simulated at 300 K to get direct Li diffusion and ionic conductivity. MSD and diffusion coefficients were extracted at 340, 360 and 380 K for 80 ps simulations time for $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ (Fig. 5b). Extrapolated room temperature ionic conductivity ($43.3 \pm 3.3 \text{ mS cm}^{-1}$) was obtained by Arrhenius plot (Fig. 5c) with an activation energy of $0.25 \pm 0.1 \text{ eV}$. During heating at 1000 K the structure was completely melted with a high rate of diffusion of each element (e.g., Cl MSD plotted in Supplementary Fig. 13a). Although amorphous phase is obtained after cooling at 1–300 K, however, Cl moved substantially with Li diffusion, while Zr and O moved slightly during AIMD simulations by extracting MSD of each atom at 380 K (Supplementary Fig. 13b). These results highlight that the deficient $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ may have Cl diffusion which may need further in-depth analyses.

To check the effect of cell size and large number of atoms on the transport mechanism and Li ionic conductivity in an amorphous structure, a supercell of $2 \times 1 \times 2$ with 96 atoms having lattice parameters of $a = 22.48571 \text{ Å}$, $b = 11.24285 \text{ Å}$, $c = 13.01690 \text{ Å}$ is used as a case study with same computational techniques with different temperature ranges and simulation time to melt the structure at 2000 K (for more detail see Supplementary note 3). The 96 atoms model exhibits fast Li ion diffusion at 300 K with Li-ion diffusion coefficients of $7.92 \times 10^{-6} \text{ cm}^2/\text{s}$ and smoother than 48 atoms model whose diffusion coefficient is $7.92 \times 10^{-6} \text{ cm}^2/\text{s}$ (Supplementary Fig. 14). The structural configuration at different temperatures for both models is presented in Supplementary Figs. (15–17) during the formation of amorphous phase. Through an extended simulation, it has been noted that the MSD of Cl is decreasing, suggesting that Cl is not actively involved in the diffusion process as a chloride ion. Rather, the motion of the Cl atom seems to be confined to rotation and oscillation around its periodic and mean position (Supplementary Fig. 18).

To confirm the formation mechanism of the amorphous phase, PCF $g(r)$ of $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ was extracted at melting, cooling and 340 K temperature. The PCF of Li–Li pair at 1000 K (dark yellow line) indicates hyperbolic shape and no significant straight peak, which indicates that the structure is completely melted (Fig. 5d). After cooling, Li pair have different

peaks indicating strong bonding length of Li with other atoms (for detail see Supplementary Fig. 19), while at 340 K Li pair appears with first peak at 3.97 Å and second high peak at radius of 6.15 Å, indicating fast diffusion and loosely bonding of Li. The inset Figure illustrates that the bonding distance between two Li atoms aligns with the PCF first peak, while the bonding length between Li and Cl falls within the range of 2.30–2.72 Å. Figure 5e shows that Li–Cl pair has the same bonding radius at each temperature (melting and cooling), with the first highest peak at the radius of $\approx 2.5 \text{ Å}$, that can be seen in the inset Figure where bonding length between Li and Cl varies in the range 2.39–2.74 Å, however the peak height decreases with temperature, showing Cl may have diffusion along Li.

Discussion

To confirm the atomic coordination of atoms with each other during phase formation of amorphous phase, Li and Zr were considered as a case study with three different stages like initial structure, melted structure and structure obtained after cooling during AIMD simulations. There is significant change in the position, relax states and coordination of all atoms after heating at 1000 K as compared to initial structure (Fig. 6). Two random Li7 and Li8 were chosen which are coordinated with six different Cl atoms (Fig. 6a). After melting the structure at 1000 K, breaking of Li–Cl coordination is observed, Li7 is coordinated with three Cl atoms and one oxygen atom, whereas Li8 coordinated with four Cl atoms (Fig. 6c) upon cooling. Initially, Zr1 and Zr2 are coordinated with five Cl and one O atom (Fig. 6d), after melting (Fig. 6e) and cooling (Fig. 6f) O1 atom acts as a bridge, connecting Zr1 and Zr2 through a strong bond (Zr1–O1–Zr2). The shuffling, movement, and weak coordination of Li with Cl during AIMD simulation process leads to fast Li diffusion, which is like the MSD results.

To further in-depth confirmation of the diffusion and migration mechanisms, a 50 ps time propagation of Li, Zr, Cl and O bond distances from the center of mass is performed at 340 K during AIMD simulations (Fig. 7). All the Li (Fig. 7a) and some Cl atoms (Fig. 7b) showed continuous motion throughout the structure during 50 ps, however some Cl atoms

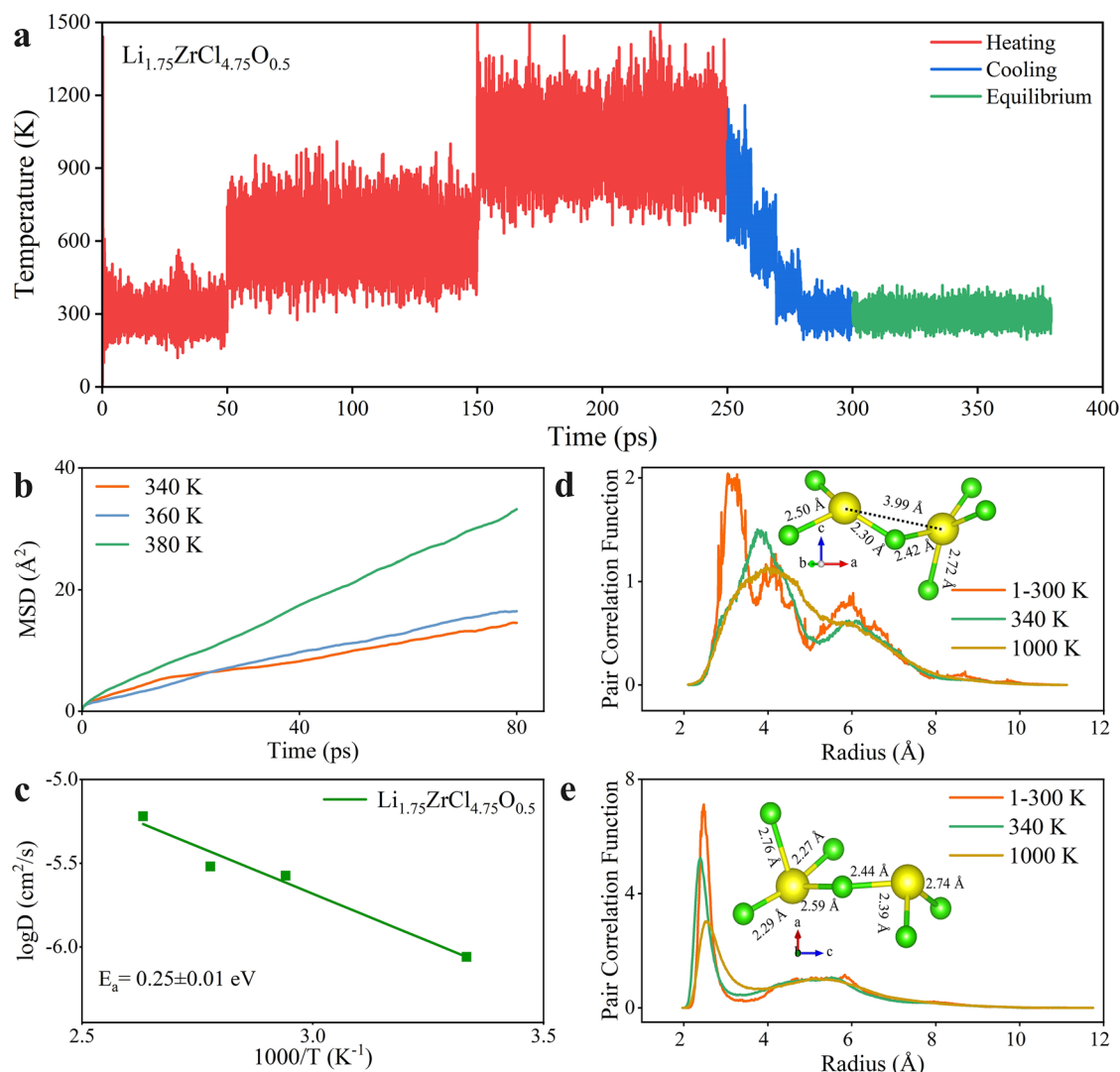


Fig. 5 | Thermal cycle, MSD, Arrhenius plots and pair correlation function for amorphous structure. **a** Thermal cycle indicating heat-quench method for the formation of amorphous structure for $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$. The red part indicates the heating process to completely melt the structure at 1000 K, blue portion represents cooling with constant simulation time, and green line indicates the final equilibrated structure at 300 K. **b** MSD of Li extracted for amorphous $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ at

different temperatures, **c** Arrhenius plots of diffusivity at different temperatures (340, 360, and 380 K) to compute extrapolated Li ionic conductivity at room temperature for amorphous $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$. Pair correlation function for amorphous structure extracted from AIMD simulations for **d** Li–Li pair and **e** Li–Cl pair at each temperature.

showed no movement, and simply vibrate at their mean position during AIMD simulations (Fig. 7c), similarly, Zr and O atoms also resist movement (Fig. 7d). The change in the bond length of some Cl atoms from the center of mass during 50 ps suggests motion like MSD results, which may need further analyses. Interestingly, all Li atoms participate in diffusion mechanism, leading to high Li ionic conductivity, and these findings correlate linearly with the observations obtained by MSD. It's worth noting that the measured lithium ionic conductivities for deficient oxyhalide SEs (with both crystalline and amorphous phases) in all the previous reports were given by electrochemical impedance spectra (EIS) measurement based on the assumption that the samples were single-ion conductors. However, the simulated result here shows the existence of diffusion of some Cl atoms that may lower or increase the lithium transference number.

It is surprising that most of the Cl atoms are localized to their mean position and only few Cl atoms indicating random motion (so called diffusion) during AIMD simulations. To find the answer, a deeper analysis of these Cl atoms (Cl25, Cl26, Cl27, Cl29, and Cl30 showing MSD and distance change in Fig. 7b) is performed for different simulation time at 340 K. These Cl atoms are positioned at corner and at x-axis (along horizontal

line). In general atoms occupying at lower corners and lower horizontal axis have no difference with upper corners and upper horizontal line due to periodic boundary condition. As an example, Cl25 atom after 20 ps positioned at lower left corner (Fig. 8a), after 40 ps upper right corner (Fig. 8b), while after 80 ps lower right corner (Fig. 8c). Such type of motion has no relation with diffusion and conduction mechanism, it is just a periodic motion of an atom along the upper and lower horizontal line. Similarly, Cl29 which is positioned inside the structure has not shown any diffusion after 20, 40, and 80 ps simulation time and remained localized at the mean position, however a negligible vibration is observed which again neglects the conduction mechanism as observed by the MSD. Cl probability density function is also computed to see the Cl diffusion and direction pathway if any, Fig. 8d indicates no Cl diffusion and conduction pathway during whole AIMD simulations, however the vibration of all Cl atoms is observed at the mean position. These analyses suggest that Cl is non mobile and neglects the results of MSD and change in length of some Cl atom showing diffusion mechanism. The amorphous phase with high concentration of intrinsic LiCl defect is also a single fast ion conductor with excellent Li ionic conductivity.

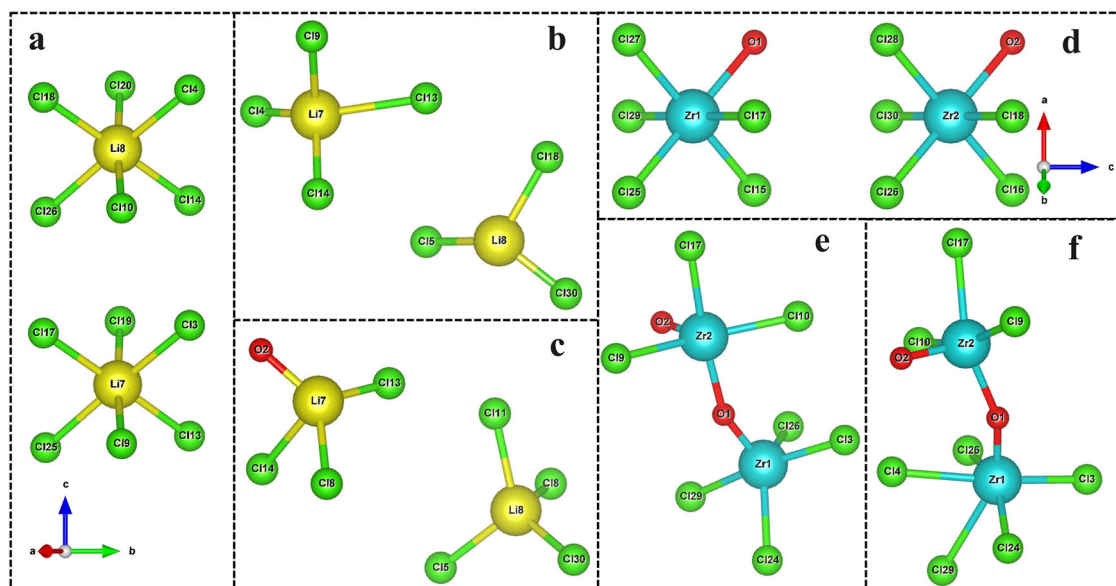


Fig. 6 | Coordination of Li and Zr. Coordination of Li7 and Li8 with other atoms **a** initial structure, **b** after melting at 1000 K, and **c** after cooling at 1–300 K. Coordination of Zr1 and Zr2 with other atoms **d** initial structure, **e** after melting at 1000 K, and **f** after cooling at 1–300 K for $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$.

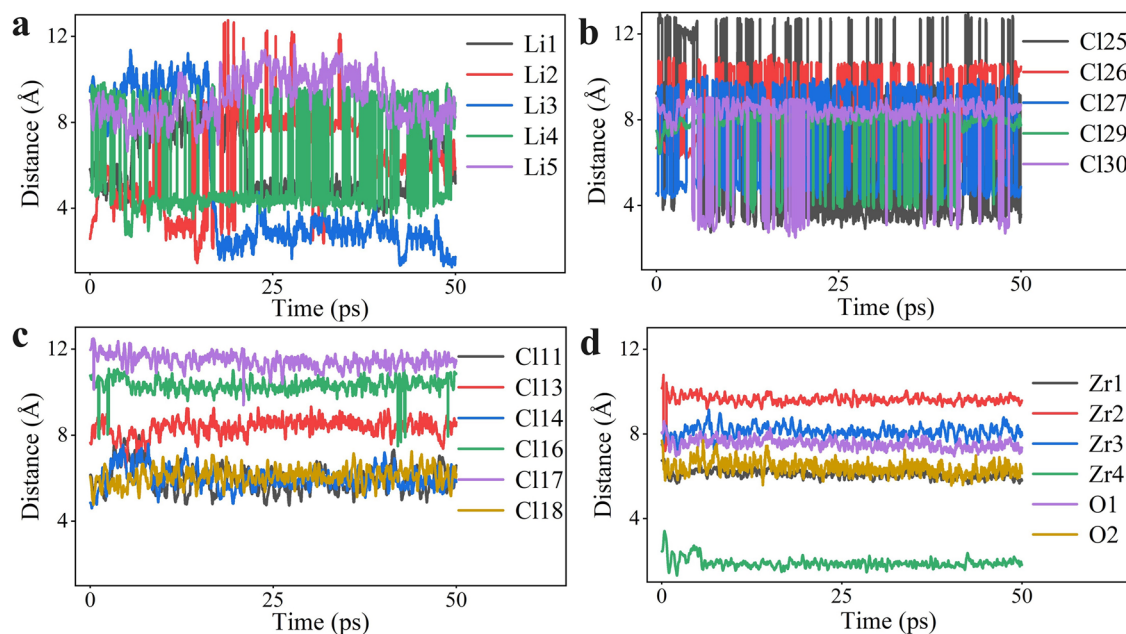


Fig. 7 | Distance from the center of mass. Distance of each atom during AIMD simulation for 0–50 ps at 340 K for **a** Li, **b** selected Cl atoms taking part in diffusion process, **c** selected Cl atoms remain localized to the mean position, and **d** Zr and O indicating no diffusion conduction in $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$.

The Nernst-Einstein equation is an approximation that neglects the ionic correlations between the trace diffusion coefficient and ionic conductivity. The ionic conductivity is a case sensitive and directly dependent on the diffusion coefficients, a small change in the diffusion coefficients values may increase/decrease an order of magnitude of the ionic conductivity. The relationship between the tracer (D^*) and charge diffusion coefficient (D) is known as Haven ratio (H_R) and can be written as^{48,49}:

$$H_R = D^*/D \quad (4)$$

The ionic correlation of crystalline solid electrolytes is extensively studied for computing the Haven ratio, however recently the ionic correlation, dynamic behavior of amorphous solid electrolytes has been analyzed in detail by computing Haven ratio^{47,50}.

The positive Haven ratio values in between the range ≈ 0.1 – 0.6 indicates the typically single ion conducting glasses with positive correlation effects^{51,52}. The $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ contains a relatively high concentration of intrinsic vacancies; therefore, the correlation may not be very strong. H_R calculations indicate (Supplementary Table 4) that the ion correlation in $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ measured is indeed notably weaker than that of other electrolytes (H_R of $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$ ⁵³ is in between 0.4 and 0.5; $\text{Li}_{10}\text{SnP}_2\text{S}_{12}$ ⁵⁴, and $\text{Li}_{10}\text{GeP}_2\text{S}_{12}$ ⁵⁰ with H_R value of 0.3). The H_R of $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is greater than 1 at 1000 K, which looks like ionic liquids⁵⁵ or molten salts⁵⁶. The H_R value at 340 K is 0.784 which is greater than 0.6 and ionic correlation is weaker but still less than 1.

To confirm the directional diffusion and trajectory⁵⁷ of Li-ion in $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and amorphous $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$, Li probability density function (PDF)⁵⁸ was extracted at 600 and 340 K during AIMD simulations.

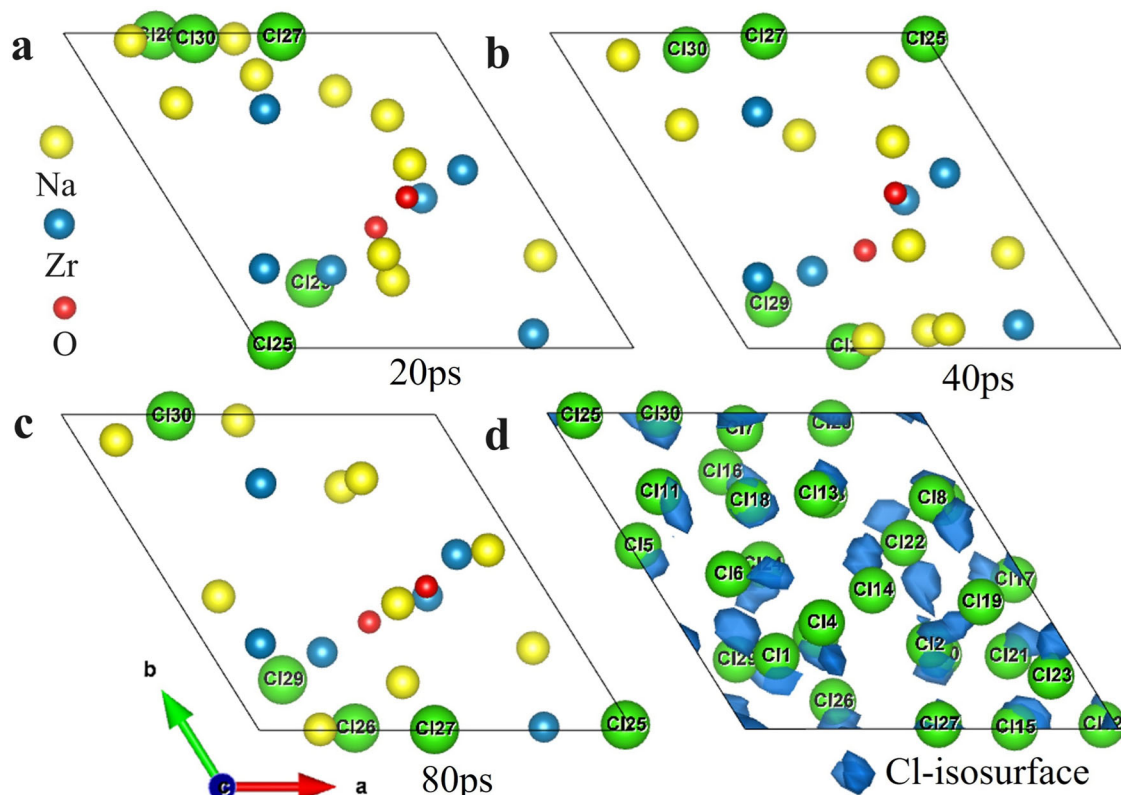


Fig. 8 | Periodic movement and Cl probability density function. Periodic movement and vibration of some Cl atoms at 340 K during AIMD simulations after **a** 20 ps, **b** 40 ps, and **c** 80 ps simulation time. **d** Cl probability density function (light blue color) with isosurface of 0.001 at 340 K for amorphous phase.

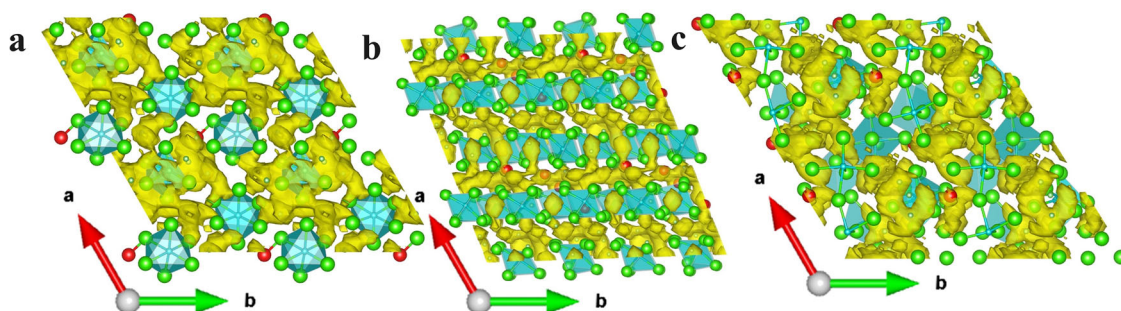


Fig. 9 | Li probability density function. Li probability density function extracted from AIMD simulations at 600 K with isosurface of 0.001 for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ for **a** $\text{P}\bar{3}\text{m}1$ phase, **b** $\text{C}2/\text{m}$ phase, and **c** Li probability density function extracted from AIMD simulations at 340 K with isosurface of 0.001 for amorphous $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$.

The isosurface (yellow in Fig. 9a, b for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ and Fig. 9c for $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$) representing the conduction path indicates three-dimensional Li diffusion throughout the structure. The 3D conduction network of Li PDF is consistent with the result obtained from MSD. The van Hove correlation function⁵⁹ was also computed for deeper understanding of the Li conduction mechanism for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ at 600 K of AIMD simulations. The distinct part of the van Hove correlation analysis for $\text{P}\bar{3}\text{m}1$ (Supplementary Fig. 13a) indicates a projecting peak near $r = 0$ after 10 ps, however $\text{C}2/\text{m}$ phase shows prominent peak near $r = 0$ after almost 1 ps (Supplementary Fig. 13b), presented in Supplementary Fig. 20a, b, which is a general behavior of correlated ion hopping mechanism⁵⁸. There is a continuous and discontinuous correlation pattern up to 30 ps, however after that Li ions exhibit a highly correlated diffusion mechanism.

In summary, $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ exhibits a theoretical lithium-ion conductivity of 1.63 and 0.24 mS cm^{-1} with an activation energy of 0.29 eV and 0.36 eV for $\text{P}\bar{3}\text{m}1$ and $\text{C}2/\text{m}$ crystalline phase, respectively. The $\text{P}\bar{3}\text{m}1$ phase is found to be energetically more stable than the $\text{C}2/\text{m}$ phase. AIMD

simulations illustrated that Zr, Cl, and O remain localized to their equilibrium position at high temperature, maintaining crystallinity of the structure for solid electrolytes. MSD and probability density function suggest, $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$ exhibits 3-dimensional conduction network and Li diffusion become highly correlated after some simulation time during AIMD simulations. However, deficient $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ (30% Li and 14% Cl depletion in $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$) is thermodynamically unstable and melts at 600 K during AIMD simulations. Simulations suggest that $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ is likely amorphous and exhibits an extraordinarily high Li ionic conductivity up to 40 mS cm^{-1} with an activation energy of 0.25 eV. The amorphous phase also suggests O acts as bridge and strongly connected with Zr atom (Zr-O-Zr) and resist to take part in diffusion mechanism. In-depth analyses of AIMD simulations suggest that the corner Cl atoms have periodic motion which may not be correlated with diffusion mechanism. The search for oxyhalide solid electrolytes could be appealing for next-generation ASSBs due to their synergistic role in recent SEs with high ionic conductivity.

Methods

Density functional theory

First-principles DFT studied were performed with Perdew–Burke–Ernzerhof (PBE) exchange–correlation functional with generalized gradient approximation (GGA)⁶⁰ using the Vienna Ab initio Simulation Package (VASP)^{61,62}. An energy cutoff of 520 eV for projector augmented wave (PAW) method⁶³, k-point $2 \times 3 \times 2$ with Monkhorst–Pack scattering method⁶⁴, to correct the underestimated bandgap by GGA, a hybrid Heyd–Scuseria–Ernzerhof (HSE06) functional⁶⁵ were used for electronic density of states (EDOS) within 10^{-2} eV/Å and 10^{-6} eV values for force and energy for convergence. The HSE06 functional is often used to compute bandgap of electrolyte materials to get an accurate upper bond limit for electrochemical window⁴⁰.

Stability and phase diagrams

Spin-polarized calculations were carried out for the relaxation of volume and atomic position in constructing ternary and quarterly phase diagram of Li–Zr–Cl–O system at 0 K. To check the phase stability of structures, ternary and quarterly phase diagrams were constructed at 0 K using convex hull⁶⁶ of the normalized energy per atom projected into the compositional coordinate space^{59,67}. The hull energy represents the stability of materials with other competitive stable phases that can be obtained by phase diagram program⁶⁸ (Pymatgen library) using pre-computed energies from the open database Material Project (MP)⁶⁹. The input parameters used for calculations are consistent with those of MP to take advantage of the large amount of calculation data available in this database. In addition to this, the hull energy computed by spin-polarized calculations is crucial in determining the stability of a structure. Structures with a hull energy of zero are known as thermodynamically stable and can easily be synthesized. On the other hand, structures with a hull energy of ≤ 25 meV fall under the category of meta-stable. Furthermore, structures with a hull energy greater than 25 meV are considered thermodynamically unstable. It is worth noting that a relatively high hull energy greater than 50 meV may pose a challenge for material synthesis. This indicates that structures with higher hull energies require more energy to synthesize and may not be as stable as structures with lower hull energies. Therefore, understanding the hull energy of a structure is essential in determining its stability to construct phase diagram and feasibility for synthesis.

Electrochemical stability

The performance of solid electrolyte could be estimated by their electrochemical window voltage with electrodes. Electrochemical stability indicates the stability of electrolytes to react with Li, in which Li acts as an open system for the electrolyte and electrode interface. In this way lithium grand potential phase diagram is constructed at different lithium potentials to compute electrochemical stability window against Li electrode. The grand lithium chemical potential phase diagrams provide information about the solid electrolyte either they are stable against Li metal anode or undergoes reduction and uptake Li at low voltage or electrolyte oxidized and lost Li at high voltage according to following equation^{40,67,70,71}:

$$\phi = E - \mu_{Li} N_{Li} \quad (5)$$

where E is the internal energy, μ_{Li} is Li chemical potential and N_{Li} is the number of Li atom present in the system. The low voltage ($\mu_{Li} \approx \mu_{Li}^0 - 5\text{eV}$) and high voltage ($\mu_{Li} \approx \mu_{Li}^0$) corresponds to cathode and anode interface, μ_{Li}^0 is the energy of metallic Li, respectively.

Ab initio molecular dynamic simulations

Ab initio molecular dynamic (AIMD) simulations under the Born–Oppenheimer approximation using Nose–Hoover thermostat⁴⁴ with canonical (NVT) ensemble were used with a time step of 2 fs with a $1 \times 1 \times 1$

k-point grid and a 400-eV energy cutoff for each temperature at 600–1000 K for 100 ps for $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$, the detail is reported in previous study^{72,73}. The properties of the material like MSD, ionic conductivity, PCF, VDOS and probability density function are analyzed from AIMD simulations. Mean square displacement (MSD) is extracted from AIMD simulations and provides the information about the mobile and immobile ions which is defined as

$$\langle [\vec{r}(t)]^2 \rangle = \frac{1}{N} \sum_{i=1}^N \langle [\vec{r}_i(t + t_0) - \vec{r}_i(t_0)]^2 \rangle \quad (6)$$

where $\vec{r}_i(t)$ is the displacement of the i -th Li ion at time t and N is the total number of Li ions. The diffusion of ions is an important property of solid electrolytes and illustrate the conduction behavior of migrated and static ions. The diffusion of an atoms is obtained by the linear fitting of MSD over time accordingly.

$$D = \lim_{t \rightarrow \infty} \left[\frac{1}{2dt} \langle [\vec{r}(t)]^2 \rangle \right] \quad (7)$$

where d is the dimension in which diffusions take place and $\vec{r}_i(t)$ is the position of ions at time t . After getting the diffusion coefficients at different temperatures activation energy ΔE_a is calculated by linear fitting of an Arrhenius plot between diffusivity and temperature ($\log \text{of } D \text{ vs } 1/T$) as follows

$$D = D_0 \exp \left(-\frac{\Delta E_a}{kT} \right) \quad (8)$$

where k is known as Boltzmann constant, T is temperature and D_0 is the diffusivity at time T . The extrapolated room temperature ion conductivity can be determined by using the Nernst–Einstein relation⁷⁴

$$\Lambda = \frac{\rho F^2 z^2}{RT} D \quad (9)$$

where ρ is called the molar density of mobile ions, z stands for charge of mobile ions, F is known as the Faraday constant, and R for the gas constant.

Additionally, AIMD simulations were performed at 1000 K using NVT ensemble for the $\text{Li}_{1.75}\text{ZrCl}_{4.75}\text{O}_{0.5}$ composition to fully melt the structure and cooled down to 300 K by heat-quenched method with constant cooling rate to obtain amorphous structure. After getting amorphous structure, further AIMD simulations were performed at 340, 360, and 380 K to compute extrapolated room temperature ionic conductivity and changes in the ionic conductivity.

The Nernst–Einstein relation, commonly used to estimate ion conductivity, has been found to be lacking in precision when it comes to capturing the complex relationship between diffusion coefficients and ionic conductivity. In order to address this issue, the concept of the Heaven ratio^{48,49} has been introduced as a means of quantifying the correlation between the trajectories of various atoms. Electrolytes can also be identified as single-ion conductors, multiple-ion conductors, and molten salts using numerical value of the Heaven ratio^{47,50}.

Probability density distribution function

Probability density distribution function $P(\mathbf{r})$ ⁵⁸ is a power tool that emphasis on the paths and direction of ions in which they diffuse and act as a charge carrier and provides the pathways for ion transportation from AIMD simulations⁵⁷ $P(\mathbf{r})$ can be define as

$$\int_{\Omega} P d\mathbf{r} = 1 \quad (10)$$

P is normalized and Ω is called the volume of unit cell.

Supplementary information

Simulated XRD pattern, Electrostatic Coulomb energy, crystallographic information of $\text{Li}_{2.5}\text{ZrCl}_{5.5}\text{O}_{0.5}$, Total Energy, Hull Energy, Optimal lattice parameter, Electrochemical and Electronic density of states, Total and directional MSD, Discussion on amorphous structure, Thermal cycle using heat-quench method, Heaven Ratio, PCF, VDOS, and distinct part of van Hove correlation function.

Data availability

The authors declare that the data supporting the findings of this study are available within the article and its supplementary information files or from the corresponding authors on reasonable request.

Received: 31 January 2024; Accepted: 28 June 2024;

Published online: 14 July 2024

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Acknowledgements

This work was supported by the Zhejiang Provincial Natural Science Foundation of China under Grant No. LY23B030003, the Natural Science Foundation of Ningbo under Grant No. 2023J200, and the Shenzhen Science and Technology Program under Grant No. KQTD20200820113047086. The work was also supported by the Laboratory of Electrochemical Energy Storage Technologies, Academy for Advanced Interdisciplinary Studies (SUSTech), the Major Science and Technology Infrastructure Project of Material Genome Big-science Facilities Platform supported by Municipal Development and Reform Commission of Shenzhen, and Center for Computational Science and Engineering at Southern University of Science and Technology.

Author contributions

F.H. and W.X. conceived the idea and performed the simulations; F.H., Y.Z., and W.X. wrote the manuscript; W.X. and J.Z. supervised and funded the work. All authors approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41524-024-01346-y>.

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